

Answer Key: Why We Age Worksheet

Teacher's copy — do not distribute to students

Section	Questions	Marks	Notes
Part A — Heading Matching	27–32	6 marks	1 mark each
Part B — Note Completion	33–36	4 marks	1 mark each; max 2 words per answer
Part C — Yes / No / Not Given	37–40	4 marks	1 mark each
Part D — Vocabulary Matching	41–48	8 marks	1 mark each
TOTAL		22 marks	

PART A — HEADING MATCHING [6 marks]

Accept only the Roman numeral. No partial marks. Example answer: Paragraph A = v (given in the question).

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|-----|--|---|
| 27. | ix — Fundamental differences in ageing of objects and organisms | Paragraph B contrasts static closed systems (objects) with open dynamic systems (organisms). |
| 28. | ii — Why dying is beneficial | Paragraph C argues that immortality would disturb evolution — death is necessary for new life. |
| 29. | vii — A stable life span despite improvements | Paragraph D notes that despite medical advances, the upper limit of human life remains ~80 years. |
| 30. | i — The biological clock | Paragraph E proposes an 'internal clock' that measures and controls the ageing process. |
| 31. | viii — Energy consumption | Paragraph F discusses metabolic rate, hibernation, diet, and gender differences in energy use. |
| 32. | iv — Prolonging your life | Paragraph G gives practical advice: relaxation, sleep, equable personality = longer life. |

PART B — NOTE COMPLETION [4 marks]

Accept only words taken directly from the passage. Maximum TWO WORDS per answer. Do not accept synonyms or paraphrases.

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|-----|---------------------------|---|
| 33. | physical chemistry | Paragraph B: 'Ageing ... must occur according to the laws of physical chemistry and of thermodynamics.' |
| 34. | thermodynamics | Paragraph B: as above. Accept either order for 33 and 34. |
| 35. | adapt | Paragraph C: 'in the course of their individual lives they are tested for optimal or better adaptation' — accept 'adapt'. |

36. **immortality**

Paragraph C: 'Immortality would disturb this system — it needs room for new and better life.'

PART C — YES / NO / NOT GIVEN [4 marks]

Note: these ask about the writer's **VIEW**S, not just factual information. Accept YES / NO / NOT GIVEN or Y / N / NG.

37. **NO**

The writer explicitly argues the opposite: wear and tear applies to objects, **NOT** to organisms, which can renew themselves.

38. **YES**

Paragraph B: 'At least as long as a biological system has the ability to renew itself it could actually become older without ageing.'

39. **NOT GIVEN**

The passage mentions continuous renewal but gives no specific percentage or time frame for body replacement.

40. **YES**

Paragraph G: 'sparing use of energy reserves should tend to extend life.' Also supported by Paragraph F examples.

PART D — VOCABULARY MATCHING [8 marks]

41. **d — wear and tear**

Gradual damage caused by continuous use over time.

42. **e — thermodynamics**

Branch of physics dealing with heat, energy and their transformations.

43. **f — dynamic equilibrium**

A stable state maintained by continuous opposing processes of equal force.

44. **c — mutation**

A change in genetic material that may produce new characteristics.

45. **a — metabolic rate**

The rate at which an organism converts food into energy.

46. **g — hibernation**

A state of deep sleep adopted to conserve energy in winter.

47. **h — immortality**

The condition of living forever without dying.

48. **b — equable**

The ability to remain calm and untroubled; emotional steadiness.

MARKING SUMMARY

Part A (Heading matching): 6 marks · Part B (Note completion): 4 marks · Part C (Y/N/NG): 4 marks · Part D (Vocabulary matching): 8 marks · TOTAL: 22 marks